

Multiple Choice (10 marks)

1. If N is the number of instances in the training dataset, nearest neighbors has a classification run time of
 - (a) $O(1)$
 - (b) $O(N)$
 - (c) $O(\log N)$
 - (d) $O(N^2)$
2. After applying a regularization penalty in linear regression, you find that some of the coefficients of w are zeroed out. Which of the following penalties might have been used?
 - (a) L_0 norm
 - (b) L_1 norm
 - (c) L_2 norm
 - (d) either (a) or (b)
3. As the number of training examples goes to infinity, your model trained on that data will have:
 - (a) Lower variance
 - (b) Higher variance
 - (c) Same variance
 - (d) None of the above
4. Which image data augmentation is most common for natural images?
 - (a) random crop and horizontal flip
 - (b) random crop and vertical flip
 - (c) posterization
 - (d) dithering
5. Which of the following is more appropriate to do feature selection?
 - (a) Ridge
 - (b) Lasso
 - (c) both (a) and (b)
 - (d) neither (a) nor (b)

True / False (10 marks)

Answer True or False

6. True or False: Regression analysis can definitively establish causal relationships between input and output variables.
7. True or False: $P(H|E, F)$ can be calculated with $P(E, F)$, $P(H)$, and $P(E|H)$, without needing $P(E, F|H)$.
8. True or False: DCGANs rely on self-attention mechanisms to enhance the stability of their training process.
9. True or False: Clustering is the most suitable method for predicting the amount of rainfall based on various cues from meteorological data.
10. True or False: Overfitting is more likely when the hypothesis space is small, as it may not capture the complexity of the data.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the index of the first occurrence of a given number in a sorted array.
12. Write a function to find minimum of two numbers.

End of Paper